

Mobile Phone Password Reset

Java Assignment: Encapsulation & State Management

The Scenario

A user has a mobile phone with an existing password. The phone should allow the user to change the password **only after entering the correct existing password**.

- The program must store the user's password as an **instance field**.
- The user enters their old password to verify their identity.
- If it matches, the new password replaces the old one.
- If it does not match, the password remains unchanged.

Examples

Example 1: Match

Existing Password: 1234

Entered Password: 1234

New Password: 5678

Password changed successfully.

Example 2: Mismatch

Existing Password: 1234

Entered Password: 1111

New Password: 5678

Incorrect password.
Password was not changed.

Required Test Cases

Test ID	Existing Password	Entered Password	New Password	Expected Result
TC01	1234	1234	5678	Password changed
TC02	1234	1111	5678	Incorrect password

Test ID	Existing Password	Entered Password	New Password	Expected Result
TC03	5678	5678	9999	Password changed
TC04	5678	1234	9999	Incorrect password
TC05	9999	9999	1234	Password changed

Your Tasks

Write a simple Java program that includes:

- A `Phone` class.
- An **instance field** to securely store the current password.
- A **method** to verify the existing password.
- A **method** to change the password after successful verification.